Master's Thesis Proposal

Title: AI Transparency in the Wild: Bridging Technical Provenance and Legal Compliance under the EU AI Act on Social Media Platforms

1. Abstract

The unchecked proliferation of Generative Artificial Intelligence (AI) has introduced profound challenges regarding digital information integrity. On social media platforms, specifically Facebook, the rapid dissemination of synthetic text and media exacerbates risks of manipulation and consumer deception. In response, the European Union has established a pioneering legal mandate: Article 50 of the EU AI Act. This directive enforces strict transparency obligations, necessitating that AI providers ensure content is machine-readable and detectable, while deployers must explicitly disclose deepfakes and AI-generated public interest texts. However, a significant technical enforcement gap persists in the wild. Real-world validation is hindered by aggressive platform-level metadata stripping, lossy image compression, and a fundamental lack of client-side auditing capabilities.

This thesis proposes a bipartite methodology, maintaining a strict 50/50 equilibrium between legal-regulatory analysis and technical software engineering. The core objective is to critically evaluate current technical standards—specifically cryptographically signed metadata (C2PA) for images and invisible statistical text watermarking paradigms (Anthropic's SynthID-Text)—and synthesize these mechanisms with the EU AI Act's disclosure guidelines. As a practical contribution, this research will culminate in the development of an "AI Transparency & Provenance Validator" prototype. This automated social media auditing pipeline will bridge the gap between theoretical legal mandates and applied cryptographic watermarking, providing a robust framework to enforce transparency and protect digital consumers.

2. How to Approach (Working Plan & Methodology)

State-of-the-Art Analysis (Guidelines & Standards): The research commences with a rigorous, dual-track review of legislative frameworks and technical provenance mechanisms. The legal analysis will decode the compliance thresholds established by EU AI Act Article 50 and the Code of Practice on Transparency, delineating the distinct responsibilities of AI providers versus deployers. Concurrently, the technical analysis will evaluate state-of-the-art provenance standards. This includes examining C2PA manifest verification for visual media and analyzing invisible text watermarking techniques. Specifically, the thesis will dissect statistical watermarking approaches—such as the SynthID-Text algorithm employed by Anthropic's Claude—which modulate the

pseudo-random token-sampling process during generation to embed machine-readable statistical signatures without degrading output quality.

The Social Media Challenge (The Facebook Context): Social media platforms present highly volatile, adversarial environments for provenance tracking. Using Facebook as the primary analytical context, this phase establishes an empirical evaluation of provenance survivability across lossy digital pipelines. The research will test the resilience of C2PA credentials against Facebook's aggressive image compression and metadata scrubbing algorithms, and assess the robustness of statistical text watermarks against minor editorial alterations and text obfuscation. Furthermore, this phase will formulate actionable advisory frameworks addressing two key demographics: providing end-users with clear visual exposure warnings to mitigate deception, and offering content creators precise deployer compliance guidelines to fulfill public interest disclosure mandates.

Tool Development & Architecture (The Technical Artifact): To bridge the regulatory-technical divide, the core engineering deliverable will be the "AI Transparency & Provenance Validator." Engineered as a robust browser extension and API auditing pipeline, the tool will passively parse DOM elements within the Facebook interface. The detection engine will simultaneously extract cryptographic C2PA credentials from visual media and decode statistical text signals from textual posts. This extracted provenance data will then be dynamically cross-referenced against a deterministic compliance engine programmed with EU AI Act thresholds. Finally, the tool will render contextual UI feed overlays, translating complex regulatory requirements into immediate, actionable alerts regarding undisclosed deepfakes or artificially generated public interest content.

Evaluation & 6-Month Timeline: The final prototype will undergo rigorous empirical robustness testing, benchmarking its detection accuracy against adversarial perturbation, metadata stripping, and content obfuscation. The project execution is structured across a highly focused, six-month trajectory:

- Month 1*:* Literature & Standards Review (EU AI Act, C2PA, SynthID-Text)
- Month 2*:* Architecture Design & Facebook Use-Case Mapping
- Month 3*:* Tool Development Phase 1 (C2PA & text watermark extraction engines)
- Month 4*:* Tool Development Phase 2 (Compliance Engine & browser UI overlay)
- Month 5*:* Empirical Robustness Testing & Perturbation Benchmarking
- Month 6*:* Legal Compliance Synthesis & Final Thesis Submission